Enterprise Decision Management with the Aid of Advanced Business Intelligence and Interactive Visualization Tools

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Summary

The algorithms behind Grapheur and LIONsolver have been widely utilized by industry to solve challenging problems like knapsack, quadratic assignment, graph problems of clustering and partitioning, vehicle routing/dispatching, power distribution, industrial production/delivery, telecommunications and industrial design. For the sake of discovering the new applications to Grapheur and LIONsolver and introducing the effectiveness of the tools offered, here the advanced tools of business intelligence and interactive visualization are seen as the reliable and novel approaches to solving the complicated problems of decision making within enterprises. Making decisions within enterprise with today's ever increasing complexity is getting harder and harder. For modeling and further solving problems as such numerous criteria and multiple disciplines need to be considered simultaneously.

Keywords

Decision-Making, Enterprise, Grapheur, Business Intelligence

Complexity Involved in Decision Making within Enterprises

Due to the rapid growing trend of globalization which leads to organizational complexity and also internal and external uncertainties, making strategic decisions become an important challenge within enterprises[1]. Making efficient decisions are crucial factor for the success of every organization. Moreover the qualities of decisions have the fundamental influence on organizational success or failure’s mission.

According to [2], in modern organizations effective decisions are essential in order to improve the functionality. Besides, decision making is the step-by-step process which starts by identifying a problem or a matter of decision, accumulate information to achieve proper solutions and finally make an optimized choice. The main issue which decision makers facing is the limitation of time and energy and the result which might be ‘satisficing’ rather than being optimized. Therefore, making decisions involves various difficulties from collecting and classifying data to coming up with the proper solutions. As [1] argued, improvement of human abilities in order to make a reliable decisions require enterprise strategies and strategic modeling that can support managers to develop their visions without knowing complicated algorithmic details and formulas. These strategic models can create a long time value for enterprises. Generally speaking, driven models can give a better insight into better understanding, exploring and concluding proper decisions within enterprises [31].

Importance of Business Intelligence in Decision Making

Business intelligence (BI) represents the ability to improve all capabilities within enterprise and transforms them into more practical data which is known as knowledge, eventually, providing the right information for the right people, at the right stage through the right channel. As a result of this organizations facing a huge amount of data that have an important role in developing new opportunities. To compete within this ever growing market, organizations need an effective strategy to implement identified opportunities [38].

Business intelligence uses technologies to evaluate data and business processes. BI systems are also known as decision support system (DSS), since they support businesses to making better decisions and solving problems [15]. The main focus in the area of DSS is how to improve efficiency in decision-making by the help of information technology and importance of graphical user interface and decision support tools in integrating technology into decision maker’s daily decisions. As [27] argued, “Intelligence is comprised of the search for problems; design involves the development of alternatives, and choice of analyzing the alternatives and choosing one for implementation.” Hence, the emphasis is analyzing a problem and developing the proper model in order to analyze different alternatives. The aim is to make the information more usable by the help of interactive visualization tools. Two advanced tools of business intelligence and interactive visualization are Grapheur and LIONsolver that are seen as the reliable and novel approaches to solving the complicated problems of decision making within enterprises. As Roberto Battiti presents in EngineSoft International Conference 2011, LIONsolver is intelligent software to create, optimize and visualize models interactively. It’s a powerful tool for data mining, modeling, problem solving and decision making in order to improve business and engineering processes. LION stand for “learning, modeling, problem-solving and optimization”. LIONsolver is the multi-purpose tool that can solve and manage enterprise problems by the aid of its modular structure within various fields, particularly, finance, logistics, bio-technology and engineering. As long as there is a business process and valid set of data, it’s easy to build a model with an incredible user-friendly interface of LIONsolver. LIONsolver contains (i) a dashboard area for visualizing data and monitoring measurements where all visualizations such as, pie chart, bubble charts, line plots, 7D plot and similarity maps are displayed (ii) a workbench for placing active tools and defining connections between databases, models and optimizers. Besides its ability to reuse data for predicative analysis, an important element of LIONsolver is Reactive Search Optimization (RSO) that is an effective method for solving multi-objective optimization problems. LIONsolver integrated with Grapheur, business intelligence and visualization software to give us an insight through data and processes more than just a simple visualization tasks.

Grapheur is a data mining, modeling and interactive visualization package implementing the Reactive Business Intelligence approach [2] which connects the user to the software through automated and intelligent self-tuning methods on the basis of visualization. The principles of Grapheur were originated from researches on Reactive Search Optimization [3]. The user friendly and innovative interface of provided visualization, via an interactive multi-objective optimization, facilitates the process of making tough decisions.

Grapheur is a handy and simple tool in which frees the mind from software complications and concentrates on mining the useful information in data. It puts the user in an interactive loop, rapidly reacting to first results and visualizations to direct the subsequent efforts, in order to suit the needs and preferences of the decision maker. The Reactive Search is utilized within Grapheur to integrate some machine learning techniques into search heuristics for the visualization complex optimization problems and interactive decision making accordingly. In Reactive Search for self-adaptation in an autonomic manner, Grapheur benefit from the past history of the search and the knowledge accumulated while moving in the configuration space [4].

Interactive Visualization Tools and Algorithms

Algorithms with self-tuning capabilities like RSO make life simpler for the final user. Complex problem solving does not require technical expertise but is available to a much wider community of decision makers [5]. Concerning solving the complex optimization problems the novel approach of Reactive Search Optimization (RSO) is to integrate the machine learning techniques into search heuristics. According to the original literatures [6] during a solving process the alternative solutions are tested through an online feedback loop for the optimal parameters' tuning. The strengths of RSO are associated to the human brain characteristics, such as learning from the past experience, learning on the job, rapid analysis of alternatives, ability to cope with incomplete information, quick adaptation to new situations and events. Methodologies of interest for RSO include machine learning and statistics; in particular reinforcement learning and neuro-dynamic programming, active or query learning, and neural networks and the metaphors for RSO derive mostly from the individual human experience. During the process of solving the real-world problems exploring the search space, utilizing RSO, many alternative solutions are tested and as the result adequate patterns and regularities appear. During the exploration the human brain quickly learns and drives future decisions based on the previous observations and searching alternatives. For the reason of rapidly exploiting the most promising solutions, the online machine learning techniques are inserted into the optimization engine of RSO. By the aid of inserted machine learning a set of diverse, accurate and crucial alternatives are offered to the decision maker. The complete series of solutions are generated rapidly, better and better ones are produced in the following search phases. The more the program runs, the bigger the possibility to identify excellent solutions. Here the decision maker decides when to stop searching. After the exploration of the design space, making the crucial decisions, within the multiple exciting criteria, totally depends on several factors and priorities which are not always easy to describe before starting the solution process. In this context the feedbacks from the decision maker in the preliminary exploration phase can be incorporated so that a better tuning of the parameters takes the end user preferences into account. As one of major character of RSO the final decision is up to the decision maker to be finalized to the machine. For this reason some qualitative factors of the decision making and optimization problem are not encoded into a computer program.

In the standard workflow of RSO the major issues of results' reproduction as well as the objective evaluation methods are directly addressed. Additionally an intelligent user is actively in the loop between a parametric algorithm and the problem solutions. In the general framework of RSO the algorithm selection, adaptation and integration, are done in an automated way. Therefor the decision maker would deal with the diversity of the problems; stochasticity and dynamicity more efficiently. Moreover the brain-computer interaction is simplified. This is done via learning for optimizing process which is basically the insertion of the machine learning component into the solution algorithm.

The term of intelligent optimization in RSO refers to the online and offline schemes based on the use of memory, adaptation, and incremental development of models, experimental algorithmics applied to optimization, intelligent tuning and design of heuristics. On-line and off-line strategies are complementary: in fact, even RSO methods tend to have a number of parameters that remain fixed during the search and can hence be tuned by off-line approaches. A long-term goal of RSO is clarified to be the progressive shift of the learning capabilities from the decision maker to the algorithm itself, through machine learning techniques [7].

According to [8] many problem-solving methods are characterized by free parameters to be tuned through a feedback loop that includes the user as a crucial learning component in which different options are developed and tested until acceptable results are obtained. The quality of results is not automatically transferred to different instances and the feedback loop can require a slow "trial and error" process when the algorithm has to be tuned for a specific application. In the Machine Learning community a rich variety of "design principles" is available that can be used to develop machine learning methods in the area of parameter tuning for heuristics. In this way the human intervention is eliminated by transferring intelligent expertise into the algorithm itself. In order to optimize the outcome setting the parameters and observing the outcome, a simple loop is performed where the parameters in a strategic and intelligent manner changed until a suitable solution is identified. In order to operate efficiently, RSO uses memory and intelligence, to recognize ways to improve solutions in a directed and focused manner. The integration of visualization and automated problem solving and optimization is the most advanced and exciting topic in this context. Often solution points are treated like vectors of numbers. By the aid of advanced visualization tools implemented and available via the software architecture packages the true meaning of numbers and the conveyed information are considered for better solutions.

Grapheur and LIONsolver are two implementations of RSO. During visualization in these software packages, it is possible to call user-specific routines associated with visualized items. This is intended as the starting point for interactive optimization or problem solving sessions, where the user specifies a routine to be called to get information about a specific solution. These implementations of RSO are based on a three-tier model, independent from the optimization algorithm, effective and flexible software architecture for integrating problem-solving and optimization schemes into the integrated engineering design processes and optimal design, modeling, and decision-making. The software implements a strong interface between a generic optimization algorithm and decision maker. While optimization systems produce different solutions, the decision maker is pursuing conflicting goals and trade-off policies represented on the multi-dimensional graphs. For solving problems with a high level of complexity, modeling the true nature of the problem is of importance and essential. Overall, if you have a challenging problem to solve, the source of power of intelligent optimization will help you both to define exactly what you want to accomplish and to actually compute one or more solutions. In some cases some decisions can, and maybe should, be deferred until some initial possible solutions are evaluated by an expert user.

Application Review

A number of complex optimization problems arising in widely different contexts and applications have been effectively treated by the general framework of RSO. This include the real-life applications, computer science and operations research community combinatorial tasks, applications in the area of neural networks related to machine learning and continuous optimization tasks. In the following we summarize some applications in real-life application areas. Worth mentioning, most of the applications of RSO are considered in [9] and the book of stochastic local search [10].

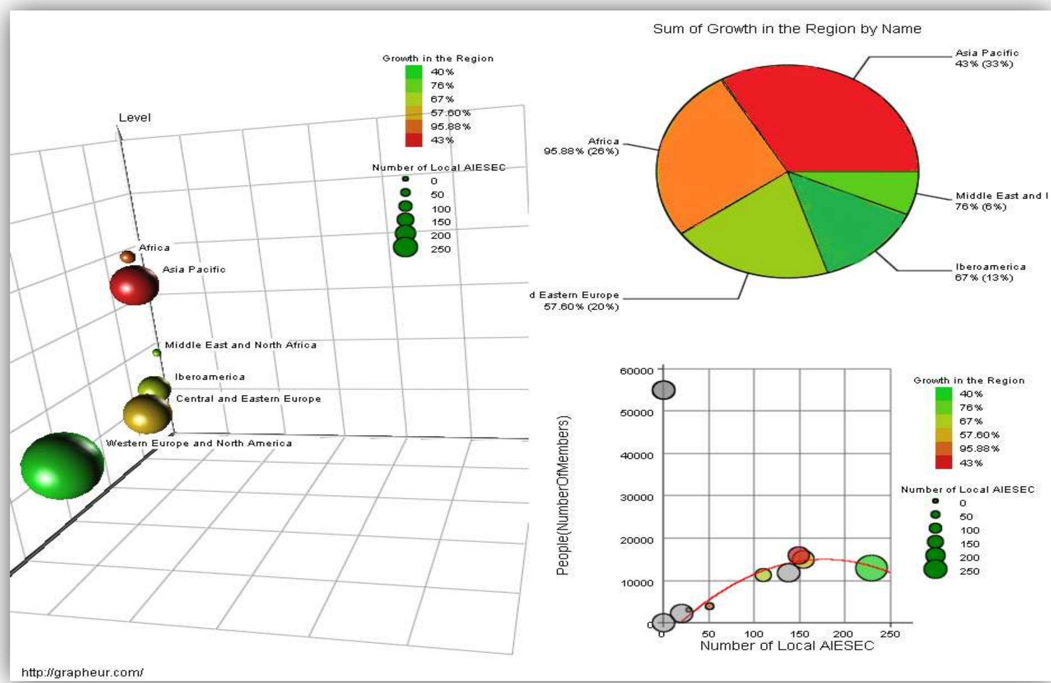
In the area of electric power distribution there have been reported a series of real-world application including service restoration in distribution systems [11], fast optimal setting for voltage control equipment [12], Service Restoration in Distribution Systems [13] and distribution load transfer operation in [14].

An open vehicle routing problem [16], as well as the pickup and delivery problem [17] both with the time and zoning constraints is modeled where the RSO methodology is applied to the distribution problem in a major metropolitan area. Alternatively to solve the vehicle routing problem with backhauls a heuristic approach based on a hybrid operation of reactive tabu search is proposed in [18]. The pattern sequencing problems in the field of industrial production planning, has been considered in [19] utilizing the modern heuristic search methods. Flexible job-shop scheduling [20] the plant location problem [21], the continuous flow-shop scheduling problem [22], adaptive self-tuning neurocontrol [23] and real-time dispatch of trams [24] were effectively solved. Moreover various applications of RSO focused on problems arising in telecommunication networks, internet and wireless in terms of optimal design, management and reliability improvements [25]. The multiple-choice multi-dimensional knapsack problem (MMKP) with applications to service level agreements and multimedia distribution is studied in [26] and [27]. In the military sector, simple versions of Reactive Tabu search are considered in [28] in a comparison of techniques dedicated to designing an unmanned aerial vehicle (UAV) routing system. Hierarchical Tabu Programming is used in [30] for finding the Underwater Vehicle Trajectories. Aerial reconnaissance simulations are the topic of [31]. Reactive Tabu Search and sensor selection in active structural acoustic control problems is considered in [32]. Visual representation of data through clustering is considered in [33]. The solution of the engineering roof truss design problem is discussed in [34].

An application of reactive tabu search for designing barreled cylinders and domes of generalized elliptical profile is studied in [35]. The cylinders and domes are optimized for their buckling resistance when loaded by static external pressure by using a structural analysis tool. Applications in bio-informatics include for example [36], which proposes an adaptive bin framework search method for a beta-sheet protein homopolymer model. Additionally a Reactive Stochastic Local Search Algorithms is used to solve the Genomic Median Problem in [37].

Case Study; Organizational chart of AIESEC

One of the usages of Grapheur within the field of enterprise management is visualizing the structure of an organization. It gives an over view of an organization's relationships and data involve. It also allows focusing on various levels in hierarchy organizational data and navigating through layers to find out the full potential within each department.
In this case we visualize an organizational chart of AIESEC. AIESEC is the world's largest student organization which providing opportunities for members to develop leadership capabilities through their internal leadership and internship programs for profit and non-profit organizations around the world. The focus of AIESEC is increasing the quality of opportunities given to its members and expanding their network.
Here we are dealing with following data;
Region: Location of AIESEC offices.
Department name: Different departments in AIESEC.
People: Number of people involves in each department.
Growth in the region: Annual growth in each region.



The network analytics module of Grapheur is designed in a way to visualized two or three dimensions. The visualization can provide more information than the traditional diagram: for instance, the color code represents the annual growth of every region, while the size of the bubbles is related to the number of people working at each region.

Conclusions

Considering the fact that complexity within enterprise is inevitable during recent years, making intelligent decisions become an important issue. Providing relevant information for the user is the crucial factor for making a decision, however the multi-objective optimization is a combined task of optimization and decision-making.
In this paper we introduced data mining and optimization tool that can support decision makers for solving problems in various areas. Such visualization tools can dramatically ease the process of decision making and therefore the management within enterprises.
In the case study we illustrate the use of Grapheur in decision making process in decision making. Considering the ability of Grapheur, the interesting patterns are automatically extracted from our raw data-set via data mining tools. Grapheur has been a facile tool in modeling the problem with the aid of a

7D plot. Once a 7D plot is created the problem could be visually analyzed from seven different perspectives simultaneously. The Graphuer appeared as effective tool for utilizing further visualization options such as similarity maps, parallel filters, and clustering would support making a confident decision

To conclude, if you have a challenging problem to solve, the source of power of intelligent optimization will help you both to define exactly what you want to accomplish and to actually compute one or more solutions. In some cases some decisions can, and maybe should, be deferred until some initial possible solutions are evaluated by an expert. Moreover, user intelligent problem solving tools can help us defining what we really want to accomplish and find a best solution available.

References

[1] Winter, T., Zimmermann, U.: Real-time dispatch of trams in storage yards. Annals of Operations Research (96), 287–315 (2000).

[2] Susan J. Miller, David J. Hickson, David C. Wilson: "Decision-Making in Organizations" (1996).

[3] Balicki, J.: Hierarchical Tabu Programming for Finding the Underwater Vehicle Trajectory. IJCSNS **7**(11), 32 (2007)

[4] Battiti Roberto and Mauro Brunato, Reactive Business Intelligence. From Data to Models to Insight, Reactive Search Srl, Italy, February 2011.

[5] Battiti, Roberto; Mauro Brunato; Franco Mascia. Reactive Search and Intelligent Optimization. Springer Verlag. (2008)

[6] Battiti, Roberto; Mauro Brunato "Grapheur: A Software Architecture for Reactive and Interactive Optimization, Proceedings Learning and Intelligent OptimizatioN LION 4, 2010, Venice, Italy (2010).

[7] Battiti, Roberto; Andrea Passerini "Brain-Computer Evolutionary Multi-Objective Optimization (BC-EMO): a genetic algorithm adapting to the decision maker." (PDF). IEEE Transactions on Evolutionary Computation (2010)

[8] Battiti, R., M. Brunato, and F. Mascia. Reactive Search and Intelligent Optimization, volume 45 of Operations research/Computer Science Interfaces. Springer Verlag, (2008).

[9] Battiti, R., Brunato, M.: Reactive search for traffic grooming in WDM networks. In: S. Palazzo (ed.) Evolutionary Trends of the Internet, IWDC2001, Taormina, Lecture Notes in Computer Science LNCS 2170, pp. 56–66. Springer-Verlag (2001)

[10] Battiti, R., Brunato, M., Delai, A.: Optimal wireless access point placement for locationdependent services. Tech. rep., Universit di Trento DIT-03-052 (2003)

[11] Błachut, J.: Tabu search optimization of externally pressurized barrels and domes. Engineering Optimization **39**(8), 899–918 (2007)

[12] Chambers, J., Barnes, J.: Reactive search for flexible job shop scheduling. Graduate program in Operations Research and Industrial Engineering, The University of Texas at Austin, Technical Report Series, ORP98-04 (1998)

[13] Chambers, J., Barnes, J.: New tabu search results for the job shop scheduling problem. The University of Texas, Austin, TX, Technical Report Series ORP96-06, Graduate Program in Operations Research and Industrial Engineering (1996)

[14] Csondes, T., Kotnyek, B., Zolt´an Szab´o, J.: Application of heuristicmethods for conformance test selection. European Journal of Operational Research **142**(1), 203–218 (2002)

[15] Consoli, S., Darby-Dowman, K., Geleijnse, G., Korst, J., Pauws, S.: Metaheuristic approaches for the quartet method of hierarchical clustering. Tech. rep., Brunel University,West London (2008)

[16] Delmaire, H., Dıaz, J., Fernandez, E., Ortega, M.: Reactive GRASP and Tabu Search based heuristics for the single source capacitated plant location problem. INFOR **37**, 194–225(1999)

[17] D. J. Power (10 March 2007). "A Brief History of Decision Support Systems, version 4.0". DSSResources.COM. Retrieved 10 July 2008.

[18] Fink, A., Voß, S.: Applications of modern heuristic search methods to pattern sequencing problems. Computers and Operations Research **26**(1), 17–34 (1999)

[19] Fink, A., Voß, S.: Solving the continuous flow-shop scheduling problem by metaheuristics.European Journal of Operational Research **151**(2), 400–414 (2003)

[20] Fukuyama, Y.: Reactive tabu search for distribution load transfer operation. In: Power Engineering Society Winter Meeting, 2000. IEEE, vol. 2 (2000)

[21] Fortz, B., , Thorup, M.: Increasing internet capacity using local search. Computational Optimization and Applications **29**(1), 13–48 (2004)

[22] Genji, T., Oomori, T., Miyazato, K., Hayashi, N., Fukuyama, Y., Co, K.: Service Restoration in Distribution Systems Aiming Higher Utilization Rate of Feeders. In: Proc. of the Fifth Metaheuristics International Conference (MIC2003) (2003)

[23] Hamza, K., Mahmoud, H., Saitou, K.: Design optimization of N-shaped roof trusses using reactive taboo search. Applied Soft Computing Journal **3**(3), 221–235 (2003)

[24] Hewage K.N., Gannoruwa A., Ruwanpura J.Y. (2011), Current Status of Factors Leading to Team Performance of On-Site Construction Professionals in Alberta Building Construction Projects, Canadian Journal of Civil Engineering.

[25] Hewage K.N., Ruwanpura J.Y., Jergeas G.F. (2009). IT Usage in Alberta’s Building Construction Projects: Current Status and Challenges, Journal of Automation in Construction.

[26] Hifi, M., Michrafy, M.: A reactive local search-based algorithm for the disjunctively constrained knapsack problem. Journal of the Operational Research Society **57**(6), 718–726 (2006)

[27] H. H. Hoos and T. Stuetzle. Stochastic Local Search: Foundations and Applications. Morgan Kaufmann, 2005.

[28] Hifi, M., Michrafy, M., Sbihi, A.: A Reactive Local Search-Based Algorithm for the Multiple-Choice Multi-Dimensional Knapsack Problem. Computational Optimization and Applications **33**(2), 271–285 (2006)

[29] J.P. Shim a,*, Merrill Warkentin a,*, James F. Courtney b, Daniel J. Power c,Ramesh Sharda d, Christer Carlsson(2002). "Past, present, and future of decision support technology"

[30] Kinney Jr, G., Hill, R., Moore, J.: Devising a quick-running heuristic for an unmanned aerialvehicle (UAV) routing system. Journal of the Operational Research Society **56**, 776–786 (2005)

[31] Kincaid, R., Laba, K.: Reactive Tabu Search and Sensor Selection inActive Structural Acoustic Control Problems. Journal of Heuristics **4**(3), 199–220 (1998)

[32] Lenne, R., Solnon, C., Stutzle, T., Tannier, E., Birattari,M.: Reactive Stochastic Local Search Algorithms for the Genomic Median Problem. LECTURE NOTES IN COMPUTER SCIENCE **4972**, 266 (2008)

[33] Linda Klebe Trevino "Ethical Decision Making in Organizations". (year) (http://www.jstor.org/stable/258313)

[34] Login, A., Areas, S.: Reactive tabu adaptive memory programming search for the vehicle routing problem with backhauls. Journal of the Operational Research Society **58**, 1630–1641(2007)

[35] Magdon-Ismail, M., Goldberg, M., Wallace, W., Siebecker, D.: Locating hidden groups in communication networks using hidden markov models. LECTURE NOTES IN COMPUTER SCIENCE pp. 126–137 (2003)

[36] Nanry, W., Wesley Barnes, J.: Solving the pickup and delivery problem with time windows using reactive tabu search. Transportation Research Part B **34**(2), 107–121 (2000)

[37] Oomori, T., Genji, T., Yura, T., Takayama, S., Watanabe, T., Fukuyama, Y., Center, T., Inc,K., Hyogo, J.: Fast optimal setting for voltage control equipment considering interconnection of distributed generators. In: Transmission and Distribution Conference and Exhibition 2002:Asia Pacific. IEEE/PES, vol. 2 (2002)

[38] Potocnik, P., Grabec, I.: Adaptive self-tuning neurocontrol. Mathematics and Computers in Simulation **51**(3-4), 201–207 (2000).

[39] Ryan, J., Bailey, T., Moore, J., Carlton, W.: Reactive tabu search in unmanned aerial reconnaissance simulations. Proceedings of the 30th conference onWinter simulation pp. 873–880 (1998)

[40] Rud , Olivia : "Business Intelligence Success Factors: Tools for Aligning Your Business in the Global Economy". Hoboken, N.J: Wiley & Sons, (year)

[41] Russell, R., Chiang, W., Zepeda, D.: Integrating multi-product production and distribution in newspaper logistics. Computers and Operations Research **35**(5), 1576–1588 (2008) Shmygelska, A., Hoos, H.: An adaptive bin framework search method for a beta-sheet protein homopolymer model. BMC Bioinformatics **8**(1), 136 (2007)

[42] V.A. Varma, G.V. Reklaitis , G.E. Blau, J.F. Pekny :"Enterprise-wide modeling & optimization—An overview of emerging research challenges and opportunities". (2007)